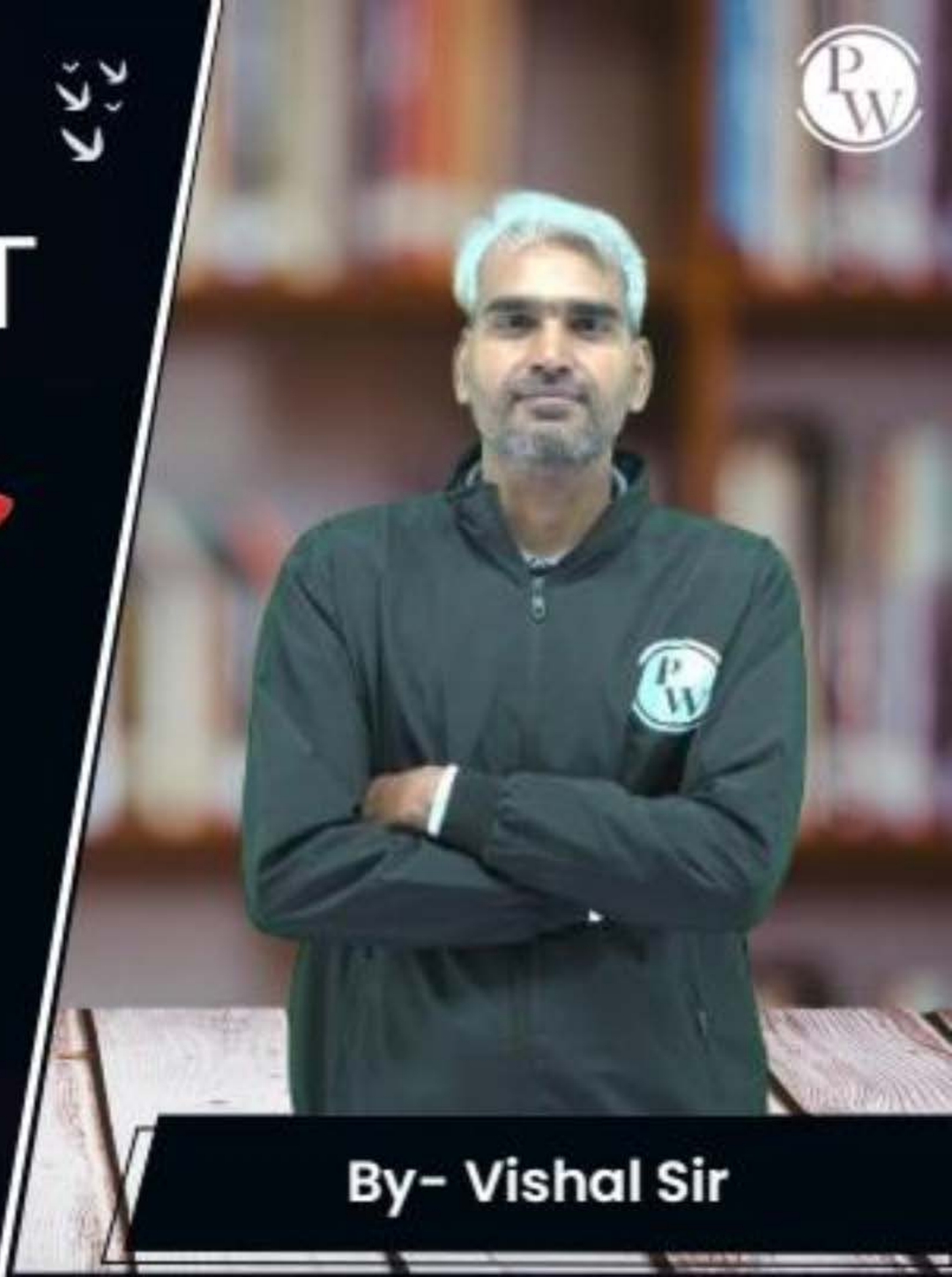
Computer Science & IT



Database Management System

Query Languages

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



Topic Practice questions on Tuple relational calculus (TRC)

Slide

Topics to be Covered



Topic	Domain relational calculus (DRC)

Slide

#Q. Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)

School (school-id, sch-name, sch-address, sch-phone)

Enrolment (school-id, sch-roll-no, erollno, examname)

ExamResult (Erollno, examname, marks)

Consider the following tuple relational calculus query

$\{ t \mid \exists E \in Enrolment\ (t = E.school\text{-}id \wedge$

$\{ x \mid x \in Enrolment \wedge x.school\text{-}id = t \wedge (\exists B \in ExamResult\ (B.erollno = x.erollno \wedge$

$B.examname = x.examname \wedge B.marks > 35 \} \mid \div$

$\{ x \mid x \in Enrolment \wedge x.school\text{-}id = t \} \mid * 100 > 35 \}$

If a student needs to score more than 35 marks to pass an exam what does the query return?

School_id Produced o/p if ∧ returns true will be as an

123 GATE 50

→ from a particular School no. of entries in which marks > 35

→ Total no. of entries in enrolment table from a particular School

Enrolment

Sch_id	Sch_rollno	Erollno	Examname
S1	1	123	GATE
S1	5	456	GATE
S1	5	456	CAT
S1	1	100	CAT
S2			
S2			
S2			

ExamResult

Erollno	Exam Name	marks
123	GATE	50 > 35 ✓
456	GATE	20 ✗ 35
456	CAT	35 ✗ 35
100	CAT	40 > 35 ✓

#Q.

A. The empty set

B. schools with more than 35% of its students enrolled in some exam or the other

C. schools with a pass percentage above 35% over all exams taken together

D. schools with a pass percentage above 35% over each exam

#Q. Consider a database that has the relation schemas
EMP(EmpId, EmpName, DeptId) and
DEPT(DeptName, DeptId).
Note that the DeptId can be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

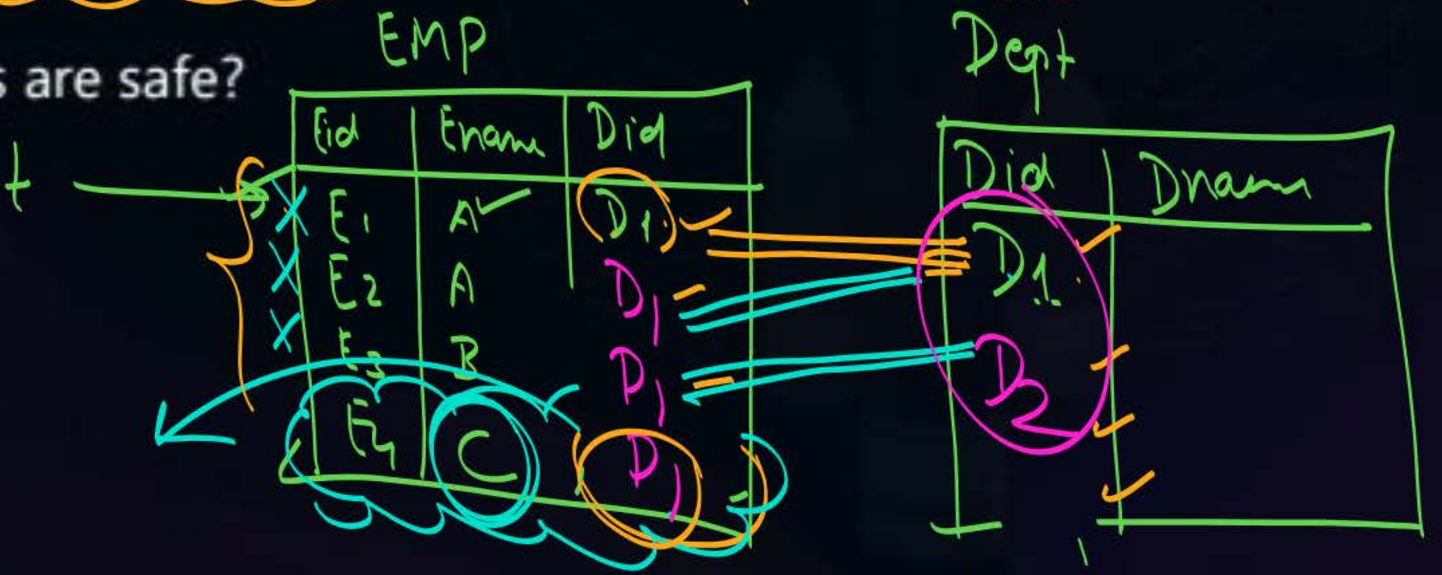
I. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \forall v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

II. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

III. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] = DeptId]))\}$

Which of the above queries are safe?

A. I and II only
B. I and III only
C. II and III only
D. I, II and III





Topic : Syntax of DRC

Domain relational Calculus

Syntax of TRC => $\{t \mid t \in Rel\}$ (t: output tuple variable)

- Each query is an expression of the form

  $\{\langle x1, x2, \ldots, xn \rangle \mid P(x1, x2, \ldots, xn)\}$

  (<x1,x2,....,xn>: output domain variable; P(x1,x2,.....,xn): Predicate formula; P(x1,x2,.....,xn): output)

- Where x1,x2,.....,xn represent domain variables.
- P(x1,x2,.....,xn) is the predicate formula.

Q: Consider the following relational schema:

Employee (Eid, Name, Salary)

Retrieve all the tuples from Employee table

TRC: $\{t \mid t \in \text{Employee}\}$

DRC: $\{\langle E, N, S\rangle \mid \langle E, N, S\rangle \in \text{Employee}\}$

To produce Eid

To produce Name

To produce Salary

#Q. Consider the following relational schemas
Employee(EmpId, EmpName, salary)

Select distinct EmpName from Employee table where salary > 2000

TRC : $\{ t.\text{EmpName} \mid t \in \text{Employee} \wedge t.\text{Salary} > 2000 \}$

DRC : $\{ \langle N \rangle \mid \langle E, N, S \rangle \in \text{Employee} \wedge \langle S \rangle > 2000 \}$

#Q. Consider the following relational schemas
Student(Rollno, name, marks)

Select distinct name and marks of the student from Student table where Rollno = 10

DRC : $\{ \langle N, M \rangle \mid \langle R, N, M \rangle \in \text{Student} \wedge \langle R \rangle = 10 \}$

(or)

$\{ \langle N, M \rangle \mid \langle 10, N, M \rangle \in \text{Student} \}$



Topic : Unsafe DRC

Unsafe TRC : $\{ t \mid t \notin \text{Relation} \}$

Unsafe DRC : $\{ \langle \text{output domain variables} \rangle \mid \langle \text{variables w.r.t Relational schema} \rangle \notin \text{Relation} \}$

#Q. Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?

"Find the distinct names of all students who score more than 90% in the course numbered 107"

```
Select distinct S.Sname
    From Student S, Regi- R
    Where (S.rollno = R.rollno AND R.Course No = 107 AND R.Percent > 90)
```

I. SELECT DISTINCT S.sname FROM Students as S, Registration as R WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90

II. $\prod_{sname}(\sigma_{courseno=107 \wedge percent > 90}(Registration \bowtie Students))$

III. $\{T \mid \exists S \in Students, \exists R \in Registration (S.rollno=R.rollno \wedge R.courseno=107 \wedge R.percent > 90 \wedge T.sname=S.sname)\}$

IV. $\{< S_N > \mid \exists S_R \exists R_P (< S_R, S_N > \in Students \wedge < S_R, 107, R_P > \in Registration \wedge R_P > 90)\}$

$< R_R, R_C, R_P > \in Reg. \wedge (< S_R > = < R_R > \wedge < R_C > = 107 \wedge < R_P > > 90$

S_R 107

A. I, II, III and IV

B. I, II and III only

C. I, II and IV only

D. II, III and IV only

* Power of TRC ≡ Power of DRC

* Power of TRC ≢ Power of basic R.A { bcoz of Unsafe TRC }

* Power of DRC ≢ Power of basic R.A { bcoz of Unsafe DRC }

* Power of Safe TRC ≡ Power of Safe DRC ≡ Power of basic R.A.

* Power of SQL ≢ Power of basic R.A.

Group by clause & aggregate functions
Can not be implemented using basic R.A.



2 mins Summary

Topic Domain relational calculus

Slide

Done with Query languages

THANK - YOU

Slide